# Tutorial Lecture 2.1 2.2

- Covered in this tutorial:
- Time Domain vs. Frequency Domain
- Continuous-Time Fourier Series (CTFS)
- Continuous-Time Fourier Transform (CTFT)
- Discrete-Time Fourier Transform (DTFT)
- Laplace Transform
- z-Transform

# Time domain and frequency domain properties

- Are the following most easily seen in time or frequency domain?
- Duration of a signal
- Peaks within a signal
- Musical notes present in an audio signal
- Frequencies most common in a signal

# What is being described here?

- A continuous periodic signal is represented as a sum of sin and cosine waves.

- A continuous signal is represented as an integral over weighted frequency components.

- A continuous signal is transformed into the complex frequency domain.

# Fourier Family

- Fourier Series (FS), Fourier Transform (FT), Discrete Time Fourier Transform (DTFT), and Discrete Fourier Transform (DFT). Which to use?
- Continuous periodic signal
- Discrete aperiodic signal
- Continuous aperiodic signal
- Discrete periodic signal

# Time shift property of Fourier Transform

- $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) \left(\cos \omega t - j \sin \omega t\right) dt$
- What is the time-shift property (in words)
- What is the time-shift property (as an equation)
- How can we prove the time shift property

# Time shift property of Fourier Transform

- How can we prove the time shift property?

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = X(\omega)$$

- New signal shifted in time $y(t) = x(t-a)$

# Z-transform

- The z-transform of a sequence $x[n]$ is given by

$$X(z) = Z(x[n]) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Which are true properties of the z-transform?

- Linear $Z(ax_1[n] + bx_2[n]) = aZ(x_1[n]) + bZ(x_2[n])$
- Multiplies by convolution $Z(x_1[n] * x_2[n]) = Z(x_1[n])Z(x_2[n])$
- Even function Z(x[n])=Z(x[-n])

# Z transform example

- Consider the positive power series: $x[n] = \begin{cases} a^n & n \geq 0 \\ 0 & n < 0 \end{cases}$
- Find the z-transform?

# Laplace transform

Definition: $X(s) = \mathcal{L}\big(x(t)\big) = \int_0^\infty x(t)e^{-st}\,dt$

Give the Laplace transform of the linear function $x(t) = at$

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